

L'armoire

Abrish Fatima

09-136252-006

Physical Project

Arduino Code in C++

!! This Code was fed into Arduino IDE & Connected to Arduino using the Arduino wire !!

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

#define S0 4
#define S1 5
#define S2 6
#define S3 7
#define sensorOut 8

int redMin = 25, redMax = 120;
int greenMin = 30, greenMax = 130;
int blueMin = 28, blueMax = 125;

// We added a boolean (true/false) to flag if a color is a "Neutral"
struct ColorProfile {
    String name;
    int r;
    int g;
    int b;
    bool isNeutral;
};

// Database with Neutral Flags
ColorProfile colors[] = {
    {"White", 255, 255, 255, true},
    {"Black", 0, 0, 0, true},
    {"Gray", 128, 128, 128, true},
    {"Brown", 139, 69, 19, true},
    {"Navy", 0, 0, 128, true},
    {"Red", 255, 0, 0, false},
    {"Green", 0, 255, 0, false},
    {"Blue", 0, 0, 255, false},
}
```

```

    {"Yellow", 255, 255, 0, false},
    {"Pink", 255, 192, 203, false},
    {"Orange", 255, 165, 0, false}
};

const int numColors = sizeof(colors) / sizeof(colors[0]);

// Array to store our 3 scanned colors
ColorProfile scannedOutfit[3];

void setup() {
    Serial.begin(9600);
    pinMode(S0, OUTPUT); pinMode(S1, OUTPUT);
    pinMode(S2, OUTPUT); pinMode(S3, OUTPUT);
    pinMode(sensorOut, INPUT);

    digitalWrite(S0, HIGH); digitalWrite(S1, LOW);

    lcd.init();
    lcd.backlight();
}

void loop() {
    // Step 1: Scan 3 Items
    for (int i = 0; i < 3; i++) {
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Scan Item ");
        lcd.print(i + 1);
        lcd.print("...");
        lcd.setCursor(0, 1);
        lcd.print("Waiting 4 secs");

        // Give you 4 seconds to place the sensor on the fabric
        delay(4000);

        // Take the reading
        scannedOutfit[i] = scanAndMatchColor();

        // Show what was detected
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Detected:");
        lcd.setCursor(0, 1);
        lcd.print(scannedOutfit[i].name);
        Serial.print("Item "); Serial.print(i + 1); Serial.print(": ");
        Serial.println(scannedOutfit[i].name);
        delay(2000); // Pause to let you read it
    }

    // Step 2: Evaluate the Combination
    evaluateOutfit(scannedOutfit[0], scannedOutfit[1], scannedOutfit[2]);

    // Wait 10 seconds before starting a brand new outfit scan
    delay(10000);
}

// --- FUNCTION: Scans hardware and finds closest color ---

```

```

ColorProfile scanAndMatchColor() {
    digitalWrite(S2, LOW); digitalWrite(S3, LOW);
    int rRaw = pulseIn(sensorOut, LOW);
    digitalWrite(S2, HIGH); digitalWrite(S3, HIGH);
    int gRaw = pulseIn(sensorOut, LOW);
    digitalWrite(S2, LOW); digitalWrite(S3, HIGH);
    int bRaw = pulseIn(sensorOut, LOW);

    int r = constrain(map(rRaw, redMin, redMax, 255, 0), 0, 255);
    int g = constrain(map(gRaw, greenMin, greenMax, 255, 0), 0, 255);
    int b = constrain(map(bRaw, blueMin, blueMax, 255, 0), 0, 255);

    int bestIndex = 0;
    long minDistance = 1000000;

    for (int i = 0; i < numColors; i++) {
        long dist = pow(r - colors[i].r, 2) + pow(g - colors[i].g, 2) + pow(b -
colors[i].b, 2);
        if (dist < minDistance) {
            minDistance = dist;
            bestIndex = i;
        }
    }
    return colors[bestIndex];
}

// --- FUNCTION: The Fashion Logic Engine ---
void evaluateOutfit(ColorProfile c1, ColorProfile c2, ColorProfile c3) {
    int neutralCount = 0;
    if (c1.isNeutral) neutralCount++;
    if (c2.isNeutral) neutralCount++;
    if (c3.isNeutral) neutralCount++;

    String verdict = "";

    // Rule 1: Monochromatic (All same color)
    if (c1.name == c2.name && c2.name == c3.name) {
        verdict = "Monochromatic!";
    }
    // Rule 2: Three Neutrals
    else if (neutralCount == 3) {
        verdict = "Classic & Safe";
    }
    // Rule 3: Two Neutrals, One Vibrant
    else if (neutralCount == 2) {
        verdict = "Great Pop of Color";
    }
    // Rule 4: One Neutral, Two Vibrant
    else if (neutralCount == 1) {
        verdict = "Very Bold!";
    }
    // Rule 5: Zero Neutrals (All vibrant)
    else {
        verdict = "Warning: Clashing";
    }

    // Display Final Result
    lcd.clear();
}

```

```
    lcd.setCursor(0, 0);  
    lcd.print("Outfit Verdict:");  
    lcd.setCursor(0, 1);  
    lcd.print(verdict);  
  
    Serial.print("--- VERDICT: ");  
    Serial.println(verdict);  
    Serial.println("-----");  
}
```